

# Vo Hong Quan

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## EDUCATION

**Ho Chi Minh City University of Technology and Engineering**, Ho Chi Minh City, Vietnam Expected Dec 2026  
Bachelor of Robotics and Artificial Intelligence GPA: 3.22/4.0  
Relevant coursework: Machine Vision, Artificial Intelligence, Artificial Neural Networks, Practice of Artificial Intelligence

## SKILLS & PROFICIENCIES

**Programming:** Python, C/C++, MATLAB (Simulink, Simscape, RL Toolbox)  
**Reinforcement Learning & Control:** PPO, SAC, reward design, hyperparameter tuning, cascaded PID, LQR  
**Robotics & Simulation:** NVIDIA Isaac Sim, Isaac Lab, PhysX, ROS 2, URDF, ESP32  
**Machine Learning & Vision:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO  
**Tools:** Git, GitHub, Linux, Conda, Jupyter/Colab, pytest  
**Languages:** Vietnamese (native), English (TOEIC 715/990)

## PROFESSIONAL EXPERIENCE

**Robotics Lab, Ho Chi Minh City University of Technology and Engineering** Feb 2025 – Present  
*Research Assistant* Ho Chi Minh City, Vietnam

- Train and evaluate reinforcement learning controllers for real-time robotic systems.
- Design reward functions and tune hyperparameters for stable policies.

## PROJECT EXPERIENCE

**Crazyflie-IsaacLab: RL-Tuned PID Gains for a Crazyflie Quadrotor** Jan 2026 – Sep 2026  
*Python, PyTorch, PPO (rs\_l\_rl), NVIDIA Isaac Lab* github.com/vohongquann/Crazyflie-IsaacLab

- Built a four-layer flight controller (50–500 Hz) and physics-based Crazyflie model in Isaac Lab 3.0.
- Trained one PPO MLP per layer to tune 9 PID gains, reducing tracking error 23–62% versus PID (attitude: 15.5° to 5.9°).
- Anchored zero network output to the stable PID baseline and sized each MLP for STM32 deployment.

**BalancingRobot-IsaacLab: Self-Balancing Robot with an RL Cascade** Jun 2025 – Dec 2025  
*Python, PPO (rs\_l\_rl), ROS 2, ONNX, LQR, PID* github.com/vohongquann/BalancingRobot-IsaacLab

- Trained a PPO velocity policy that maps commands and raw IMU/encoder data to wheel torques.
- Added a frozen position policy; recorded zero falls in 768 randomized episodes and 8 mm median goal error.
- Beat LQR on speed steps: 0.026 m/s error in 0.36 s versus 0.067 m/s in 1.1 s.

**RotaryPendulum-IsaacLab: Learned Cascade PID for a Rotary Inverted Pendulum** Feb 2025 – May 2025  
*Python, PPO (rs\_l\_rl), NVIDIA Isaac Lab, PhysX, URDF* github.com/vohongquann/RotaryPendulum-IsaacLab

- Modeled the QUBE-Servo 2 from CAD with a DC motor and quantized encoders at 500 Hz.
- Trained a PPO policy on 4,096 robots in 64 minutes to tune six cascade-PID gains at 100 Hz.
- Validated on 256 randomized robots: 0.40 s swing-up, 0.23–0.50 s settling and at most 1.4° median overshoot.

## CERTIFICATIONS

TOEIC Listening & Reading: 715/990, IIG Vietnam 2025